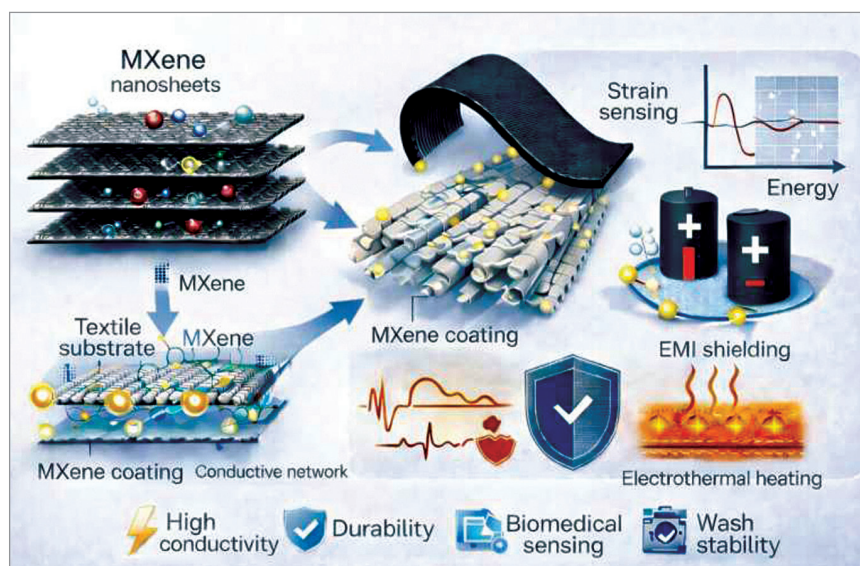


RESEARCH

& INNOVATION UPDATES

Smart textiles to power next-gen wearables

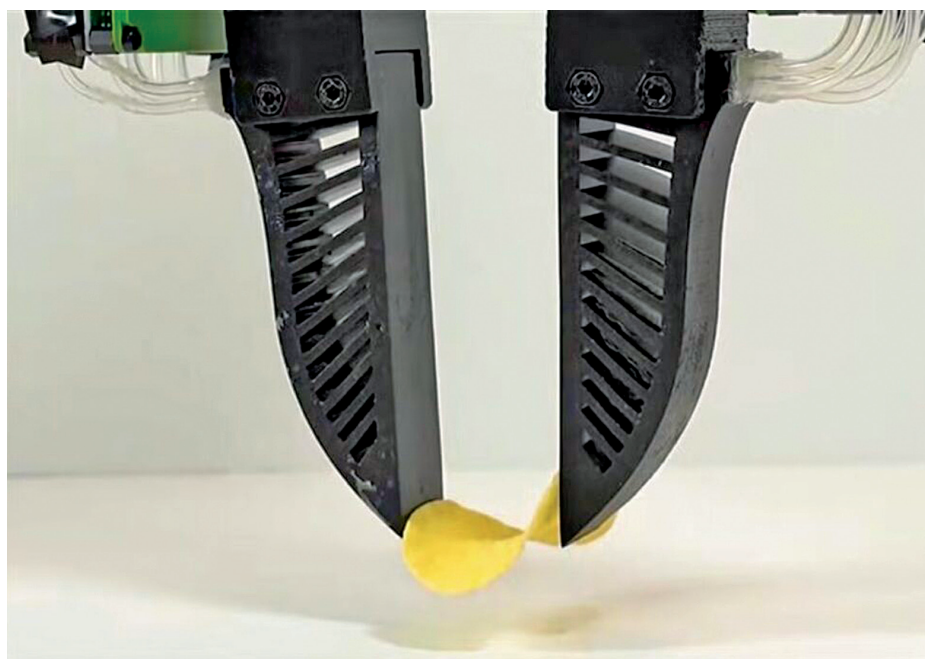


Graphical extract of MXene/Ni coating process (Credit: ACS Omega)

Researchers from the University of Georgia have developed MXene-based smart textiles that can generate power, store energy, and monitor health signals, potentially transforming wearable electronics. The fabrics integrate conductive MXene nanomaterials into conventional fibres, enabling garments to harvest solar energy and run embedded sensors without external batteries. Sensors in the textile can track vital signs such as heart rate, body temperature, and blood pressure for continuous health monitoring. The technology explored multifunctional e-textiles capable of sensing, energy harvesting, and antimicrobial protection, thereby expanding possibilities for medical garments and next-generation wearable devices.

Slip-sensing robotic grippers for delicate items

A new robotic gripper capable of detecting when objects begin to slip has been developed, enabling robots to adjust their grip in real time and handle fragile items without damage. Engineered by a team at the University of Texas at Austin in the United States, the system, called FORTE, combines soft robotic fingers with embedded tactile sensors that monitor pressure changes as the gripper contacts an object. These sensors detect early slip signals and trigger rapid adjustments in grip force. Tests showed the gripper could pick up delicate items such as potato chips and raspberries while maintaining a 91.9% single-trial grasp success rate. The system also identified 93% of slip events with perfect precision, ensuring that force is applied only when necessary. The technology could improve robotic handling in food processing, healthcare, and electronics manufacturing, where delicate manipulation is critical.



Robotic grippers so sensitive they can grab a potato chip (Credit: <https://news.utexas.edu>)